

Kuan-Chen Chen

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EDUCATION

University of Florida

Master of Science in Applied Data Science

National Sun Yat-Sen University

Bachelor of Science in Applied Mathematics

Honors & Awards: The Valedictorian of Department of Applied Mathematics, Distinguished Service Award

Gainesville, FL

8/2024 - 5/2026

Kaohsiung, Taiwan

9/2019 - 6/2023

PROJECT EXPERIENCE

cAIdron (Recipe Generation LLM) [cAIdron - Kuan-Chen Chen](#)

Gainesville, FL

- Built an AI-powered recipe generation app where users photograph ingredients and receive complete recipes with AI-generated food images; iteratively shipped two production versions, cutting end-to-end pipeline latency from 40–80s (v1) to 10–20s (v2) through architectural redesign and parallel execution.
- Developed v1 pipeline using CLIP and DETR for multi-ingredient detection across 525+ food classes (88% F1); fine-tuned Llama 3.2 1B via QLoRA on 7,913 curated RecipeNLG recipes, updating only 0.23% of parameters (2.8M) while achieving 92/100 recipe quality score.
- Rebuilt as cloud-native v2 with LangGraph StateGraph orchestration, replacing local GPU models with Groq Llama 4 Scout 17B for vision inference and Llama 3.3 70B for generation; integrated ChromaDB RAG and Tavily web search via parallel fork/join execution, eliminating 4–6 GB VRAM dependency.

Climate Change Innovation Competition, The Ministry of Education (Taiwan)

Taipei, Taiwan

- Built DARTS campus mobility prediction system by integrating multi-source APIs (TDX/PTX), enabled automated 24/7 data pipeline and real-time forecasting of pedestrian, bicycle, and bus demand
- Developed hybrid time-series model (ARIMA-LSTM) with data preprocessing and feature engineering, achieved 86.1% accuracy; delivered actionable insights adopted by university, optimizing bus schedules and reaching national finals (Top 121 teams)

WORK EXPERIENCE

Florida Gators Women's Basketball

Gainesville, FL

Data Scientist Intern

8/2025 - Present

- Built an opponent-strength-adjusted player evaluation system (OA-Rating) using XGBoost and SHAP, analyzed 120K+ game-level observations across 5,500+ NCAA players and 350+ teams
- Implemented an Elo-based team rating system ($p = 0.83$ vs. NCAA NET) to track real-time team strength and support game preparation
- Developed a production R Shiny dashboard for player comparison and opponent scouting, delivering insights that contributed to team improvement from 16–17 (NET 60) to 18–15 (NET 48)

Bioinformatics Lab, University of Florida

Gainesville, FL

Student Research Assistant

12/2024 - Present

- Designed stacking ensemble algorithm combining XGBoost, LightGBM, and Balanced Random Forest with logistic regression meta-learner to classify 10,486 T-cell receptor CDR3 sequence, improving robustness on highly imbalanced data (1:23) and achieving ROC-AUC 0.876
- Developed biologically-informed data augmentation algorithm using BLOSUM62 substitution matrix, generating conservative sequence variants to expand minority class 10x (432 $\rightarrow$ 4,622) and mitigate severe class imbalance (1:23 $\rightarrow$ 1:1.5)
- Engineered 42 physicochemical features from protein sequences and optimized decision thresholds under class imbalance, improving precision to 0.716 for high-confidence, interpretable predictions.

Optimized AI Conference

Atlanta, GA

Volunteer Staff

4/2025, 3/2026

Glory Integrated Marketing Co. Ltd

Taipei, Taiwan

Data Analyst Intern

8/2023 - 12/2023

SKILLS

- Languages: Python, SQL, R | Databases: MySQL, MongoDB, ChromaDB
- ML/DL: PyTorch, scikit-learn, LangSmith, LangGraph, HuggingFace Transformers, SHAP
- Web & Viz: Flask, React.js, Gradio, Dash, Shiny, Power BI, Tableau